Tensor Omics planning

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1 Google Slides for illustrations of figures

See here, please.

2 Relative Axes Expression (RAE) Projection

2.1 Method: Projection onto the Diagonal-Orthogonal Plane

Given gene expression vectors in an n-dimensional semantic space (e.g., tissues, conditions), we project them onto the hyperplane orthogonal to the space diagonal to analyze relative expression across axes.

Let:

- $\mathbf{x} \in \mathbb{R}^n$: the expression vector of a gene (e.g., paralog or ortholog),
- $\mathbf{d} = \frac{1}{\sqrt{n}}(1, 1, \ldots, 1)$: the unit space diagonal vector.

We define the projection of $\mathbf{x}$ onto the plane orthogonal to $\mathbf{d}$ as:

$$\mathbf{x}_\perp = \mathbf{x} - (\mathbf{x} \cdot \mathbf{d})\mathbf{d}$$

Let $\mathbf{o}_\perp$ and $\mathbf{p}_\perp$ be the projections of the ortholog centroid and the paralog outlier, respectively. Then:

$$\Delta\mathbf{v} = \mathbf{p}_\perp - \mathbf{o}_\perp \quad \text{(difference vector)}$$
$$\hat{\Delta\mathbf{v}} = \frac{\Delta\mathbf{v}}{\|\Delta\mathbf{v}\|} \quad \text{(normalized difference vector)}$$

To analyze shifts in direction, we define the **clock hand vectors**:

$$\mathbf{v}_{\text{hand}} = \mathbf{x}_\perp - \mathbf{d}_\perp$$

which point from the diagonal anchor to the projected expression vector.

The **angle** φ between clock hands (e.g., of a paralog and its ortholog) quantifies the shift in **tissue preference** or semantic emphasis.

2.2 Biological Interpretability

- **Angle** φ: Measures the rotation in tissue (or axis) preference. A change in φ indicates that the expression emphasis has shifted between semantic dimensions (e.g., from one tissue to another).
- **Difference vector** $\Delta\mathbf{v}$: Represents the raw shift in relative expression across semantic axes.

- **Normalized difference vector $\hat{\Delta\mathbf{v}}$**: Reflects the relative (percental) change in semantic emphasis, independent of expression magnitude.
- **Subfunctionalization**: Can be detected if the clock hand vectors of multiple paralogs sum approximately to the ortholog's clock hand:

$$\mathbf{v}_{\text{hand}}^{(p_1)} + \mathbf{v}_{\text{hand}}^{(p_2)} \approx \mathbf{v}_{\text{hand}}^{\text{orth}}$$

- **Neofunctionalization**: Can be inferred when a paralog expresses in a semantic dimension (e.g., tissue) where the ortholog showed no expression, observable as clock hands extending into previously unoccupied angular sectors.

2.3 Terminology

- **RAE**: Relative Axes Expression — the name for this projection-based analysis method, generalizable across omics types and semantic spaces.
- **Plane $\Pi_\perp$**: The plane orthogonal to the space diagonal $\mathbf{d}$.
- **Clock hand vector**: A vector drawn from the projected diagonal anchor to the projected expression vector.
- **Angle φ**: Captures the directional change in expression emphasis between genes.

2.4 Clock Plots for Time Trajectory Visualization

To visualize gene expression trajectories over time in semantic vector space (e.g., tissue or condition axes), we introduce **clock plots** in the plane $\Pi_\perp$ orthogonal to the space diagonal. Each clock hand represents the projected expression vector $\mathbf{x}_\perp^{(t)}$ at a specific time point t. The twelve o'clock position is defined by a reference vector, typically:

- a projected axis (e.g., the projected *leaf* or *seed* axis), or
- the direction orthogonal to the conserved mean trajectory (in divergence studies).

The clock plot offers intuitive interpretation:

- The **angle between hands** shows directional changes in semantic emphasis over time (e.g., a shift from one tissue to another).
- The **length of the hand** corresponds to the magnitude of deviation from uniform expression across axes.
- Time points can be annotated directly on the clock to visualize expression trajectories as semantic rotations.

Helper Distance-Time Plot. To complement clock plots, we employ a 2D helper plot:

- The x-axis represents time along the trajectory.
- The y-axis shows the Euclidean distance between the expression vector at time t and the projected ortholog centroid $\mathbf{o}_\perp$.

Together, these plots convey both the **directional trajectory** and the **divergence magnitude** over time—offering a concise geometric view of neofunctionalization, subfunctionalization, or dynamic tissue shifts in response to experimental conditions.

Projected Semantic Axes as Reference Frame. To enhance interpretability of clock plots, the original semantic axes (e.g., tissue, condition, or omics dimensions) are projected into the diagonal-orthogonal plane $\Pi_\perp$. These projected axes form a **warped coordinate star** radiating outward from the origin of the clock.

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ denote the n standard basis vectors (unit vectors along each semantic axis). We define their projections onto $\Pi_\perp$ as:

$$\mathbf{a}_{i,\perp} = \mathbf{a}_i - (\mathbf{a}_i \cdot \mathbf{d})\mathbf{d}$$

These projected axes are visualized as thin rays or arrows in the clock plot, forming a **star-like reference system**. Due to the off-center perspective created by the orthogonal projection, the angles between the projected axes are generally no longer 90° and become **visually warped**. This geometric warping encodes the semantic structure of the space and allows users to interpret trajectory movements in relation to the original biological meaning of the axes.

User-Defined Anchor Axis as Twelve O'Clock. To orient the plot meaningfully, a user-defined semantic axis is selected as the **twelve o'clock anchor**. Its projected vector $\mathbf{a}_{\text{anchor},\perp}$ defines the upward direction of the clock. All angle calculations (e.g., φ between time points or between paralogs and orthologs) are computed relative to this anchor axis.

This design enables intuitive biological interpretation, e.g.:

- setting *leaf* as twelve o'clock lets one interpret early/late developmental tissue shifts as clockwise or counterclockwise movement,
- selecting the direction orthogonal to the conserved reference trajectory highlights deviations from evolutionary conservation.

Together, the warped semantic star and the twelve o'clock anchor provide a geometrically faithful yet biologically meaningful visual frame for interpreting expression dynamics and functional divergence across time, gene families, or omics types.